

Primary Hypertension

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DEFINITION

Primary (essential) hypertension is defined by most international societies as an office-based blood pressure (BP) with systolic pressure of 140 mm Hg or higher or diastolic pressure of 90 mm Hg or higher,¹ but in 2017 it was defined by the American College of Cardiology/American Heart Association (ACC/AHA) as BP of 130/80 mm Hg or higher without an identifiable cause² (Table 35.1). The ACC/AHA guidelines have a more stringent definition because of the recognition of a graded increase of cardiovascular (CV) events that exist even in those with systolic BP of 120 to 129 mm Hg and less than 80 mm Hg (which is now termed *elevated BP*). An argument against the ACC/AHA guidelines is that they will lead to a majority of adults being given the diagnosis of hypertension requiring therapy.

Blood pressure is normally variable and related to a natural circadian rhythm, with the most significant increase in BP in the morning (6–10 AM). Blood pressure falls during sleep secondary to a decrease in sympathetic nervous system (SNS) tone and reduced activity of other neuroendocrine systems. There are also minute-to-minute variations in BP (Fig. 35.1). Transient elevations in BP, reaching 150 mm Hg systolic, occur in most normotensive individuals in any given day, especially during exercise.³ Thus, diagnosis of elevated BP and/or hypertension should involve multiple measurements (see Chapter 34).

When only the systolic BP (SBP) is elevated (SBP >140 and diastolic BP [DBP] <90 mm Hg), the term used is *isolated systolic hypertension*. “*White coat*” hypertension is an increase of more than 20 mm Hg in SBP noted only in the physician’s office and above that seen at home or in another setting. In contrast, *masked hypertension* is BP that is normal in the office but elevated by more than 20 mm Hg when measured by ambulatory BP measurement.

Other terms used to describe specific clinical presentations include *hypertensive emergency*, associated with acute end organ damage requiring immediate treatment, usually in a critical care setting, and *hypertensive urgency*, in which there is no acute organ damage and BP needs correction in hours or a few days (see Chapter 38). In hypertensive emergencies in which an acute decrease of kidney function occurs, the reduction of BP will halt, prevent, or reverse decreasing glomerular filtration rate (GFR). These terms have replaced “malignant hypertension” and “accelerated hypertension.” *Resistant hypertension* is defined as hypertension that remains greater than 140/90 mm Hg (or 130/80 by US guidelines) despite use of three maximally dosed antihypertensive medications of different classes, including a diuretic. Patients that require four or more medications are also considered to have resistant hypertension. The definition of resistant hypertension is arbitrary and aims to identify patients that may benefit from special diagnostic or therapeutic considerations. *Pseudoresistant* hypertension defines uncontrolled hypertension that appears resistant but is due to other causes, usually poor adherence to therapy or inappropriate

measurement (e.g., using an inappropriately small blood pressure cuff).

EPIDEMIOLOGY

Hypertension affects nearly one-third of adults worldwide, with a higher prevalence in low- and medium-income countries (31.5%, 1.04 billion people) than in high-income countries (28.5%, 349 million people).⁴ The prevalence of hypertension rose significantly in the twentieth century from approximately 5% in the United States in the early 1900s (based on studies using 140/90 mm Hg as the cutoff), possibly because of the increasing longevity of the population and the rising prevalence of obesity (Fig. 35.2).

There are multiple risk factors for primary hypertension (Table 35.2). Older age is a strong risk factor, especially after the age of 65 years, such that by 80 years of age, 90% of individuals are hypertensive.⁵ Interestingly, the age-related increase in prevalence of hypertension has not been uniformly observed in all populations (e.g., the Yanomamo Indians on their native hunter-gatherer diet). Hypertension is more common in men, although the prevalence in women is similar to and slightly exceeds that of men after age 55 years. Certain ethnic groups are at increased risk of hypertension, especially African Americans, Filipino Americans, Australian Tiwi, and the Maori of New Zealand. Other risk factors for hypertension include family history, obesity, insulin resistance, hyperuricemia and/or gout, sleep disorders including sleep apnea, and persistent high-stress environments either at work or home. Certain physical features, such as elevated heart rate or an increased BP response to exercise, are also predictive, as is elevated hematocrit.

Genetic factors play a role in 20% to 30% of patients with primary hypertension, likely from the cumulative effect of multiple susceptibility genes. To date over 90 genetic polymorphisms have been identified that are associated with primary hypertension,⁶ including polymorphisms linked with natriuretic peptide, vasoactive mediators (angiotensinogen, endothelial nitric oxide synthetase, prostacyclin synthase, β_2 -adrenoceptor, 20-HETE synthase [CYP4F2 gene], G protein β_3), mediators of vascular smooth muscle tone (calcium-dependent potassium channel, *KCNMA*), immune signaling (e.g., heat shock proteins), and mediators controlling kidney sodium transport (α -adducin and 11 β -hydroxysteroid dehydrogenase type 2, aldosterone synthase, “with no lysine” kinase [WNK] kinases). Although genetic polymorphisms involving sodium excretion are common in persons with primary hypertension,⁷ more than 80% also have effects on the immune system.⁶

Hypertension is also more likely to occur if the mother has a history of hypertension, obesity, preeclampsia, or malnutrition. A history of intrauterine growth restriction or low birthweight also are risk factors for hypertension.

TABLE 35.1 Categories of Blood Pressure in Adults

Category	SBP (mm Hg)	DBP (mm Hg)
ACC/AHA Hypertension Guidelines (2017)		
Normal	<120	and <80
Elevated	120–129	and <80
Hypertension		
Stage 1	130–139	and 80–89
Stage 2	≥140	and ≥90
European Society of Hypertension Guidelines (2018)		
Optimal	<120	and/or <80
Normal	120–129	and/or 80–84
High-normal	130–139	and/or 85–89
Grade 1 hypertension	140–159	and/or 90–99
Grade 2 hypertension	160–179	and/or 100–109
Grade 3 hypertension	≥180	and/or ≥110
Isolated systolic hypertension	≥140	and <90

ACC/AHA, American College of Cardiology/American Heart Association; DBP, diastolic blood pressure; SBP, systolic blood pressure.

Diurnal Variation in Blood Pressure

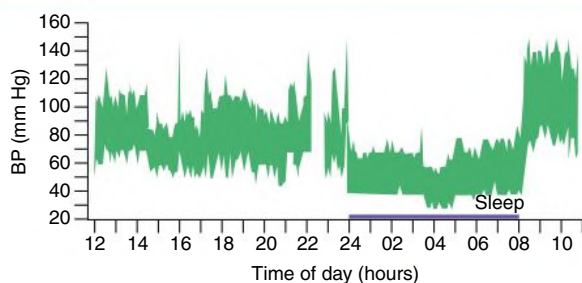


Fig. 35.1 Blood Pressure (BP) Variability in a Normotensive Individual. In most individuals, systolic blood pressure reaches 150 mm Hg at least once daily. (From Bevan AT, Honour AJ, Stott FH. Direct arterial pressure recording in unrestricted man. *Br Heart J.* 1969;31[3]:387–388.)

Epidemic of Hypertension

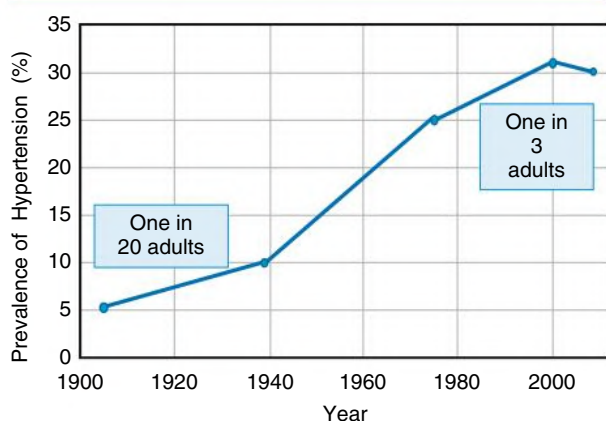


Fig. 35.2 Epidemic of Hypertension. The prevalence of primary hypertension (defined as blood pressure >140/90 mm Hg) in the United States increased from 11% in 1939 to 31% in 2008, leveling to 29% of the population in 2012.

TABLE 35.2 Major Risk Factors for Primary Hypertension

Genetic	Family history Polymorphisms (adducin, endothelial nitric oxide synthase, angiotensinogen, β_2 -adrenoceptor, human G protein β_3 subunit)
Congenital	Low birthweight, low nephron number, maternal hypertension, maternal preeclampsia, maternal malnutrition
Physical	Obesity, older age, African American, African Caribbean, some Bantu-speaking peoples in Africa, increased heart rate (>83 beats/min), increased emotional stress
Diet/toxin	Increased sodium intake, low potassium intake, low dairy products intake, heavy alcohol intake, high intake of added sugars, low level lead or cadmium intoxication
Metabolic ^a	Elevated serum uric acid, insulin resistance, elevated hematocrit
Other	Low socioeconomic status, urban versus rural residence

^aLaboratory-based parameters.

Dietary and environmental factors also contribute to the risk for hypertension. Obesity, with or without features of insulin resistance and metabolic syndrome, is a major risk factor for hypertension, and the increased incidence of obesity parallels the rise in hypertension in most countries. Epidemiologic and interventional studies have linked high salt and low potassium intake with persistent BP elevation (especially in African Americans), whereas increasing potassium intake lowers BP in both experimental and human studies. Intake of added sugars (e.g., sucrose, high-fructose corn syrup) predicts higher BP, and excessive alcohol consumption, physical inactivity, and chronic low-level exposure to lead and cadmium have all been associated with the development of hypertension.

ETIOLOGY AND PATHOGENESIS

Primary hypertension is thought to be due to a defect in kidney sodium handling. Whereas increasing salt intake will only increase BP slightly (several mm Hg) in normal individuals, in subjects with *salt-sensitive hypertension*, the BP response is shifted rightward with a change in slope, leading to a greater relative increase in BP for the same load of salt (Fig. 35.3).⁸ In contrast, some hypertensive patients, especially those younger than 40 years, are considered to have *salt-resistant hypertension*, as the response to a salt load is similar to that in normal individuals but simply shifted rightward. There is some evidence that individuals can begin with salt-resistant hypertension and evolve to being salt-sensitive.

The importance of salt in primary hypertension is supported by a direct association between dietary sodium content and the prevalence of hypertension in diverse populations, and intervention studies in which salt restriction has led to improvement in hypertension.⁹ Several hypotheses have been proposed to explain the kidney defect in sodium excretion in primary hypertension.

Genetic (Polygene) Hypothesis

Lifton and colleagues determined that many monogenic forms of both hypertension and hypotension involve genes regulating kidney sodium transport (Chapter 49).⁷ Lifton has suggested that polymorphisms in genes that favor sodium reabsorption in the kidney, coupled with excessive salt intake (>10 g/day) that is pervasive in Western diets, may underlie the pathogenesis of primary hypertension.⁷ Likewise,

Physiologic Defect in Sodium Excretion in Essential Hypertension

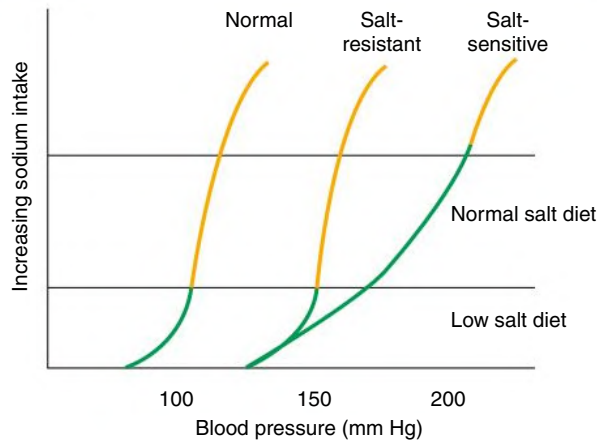


Fig. 35.3 Physiologic Defect in Sodium Handling in Primary Hypertension. Evidence suggests that increasing salt in the diet will only slightly increase blood pressure (BP) in subjects with normal BP or subjects with salt-resistant hypertension, whereas subjects with salt-sensitive hypertension will show an exaggerated BP response. (Modified from Guyton AC, Coleman TG, Cowley AV Jr, Scheel KW, Manning RD Jr, Norman RA Jr. Arterial pressure regulation. Overriding dominance of the kidneys in long-term regulation and in hypertension. *Am J Med.* 1972;52[5]:584–594.)

mutations that favor sodium excretion might provide protection from hypertension, such as polymorphisms involving the Na-K-2Cl cotransporter SLC12A1, the inward rectifier K⁺ channel KCNJ1 (carrier state for Bartter syndrome), and the Na-Cl cotransporter SLC12A3 (carrier state for Gitelman syndrome).

Nevertheless, the single-nucleotide polymorphism (SNP) variants identified by genome-wide association studies account for only 4% to 13% of the BP variability in studies of quantitative changes in BP,^{2,10} and studies that analyzed hypertension as a dichotomous trait found only two SNPs corresponding to hypertension: one associated with the uromodulin gene and the other located near the endothelial nitric oxide synthase gene.¹¹

Congenital (Low Nephron Number) Hypothesis

Environmental stress during pregnancy may cause epigenetic changes that affect the fetus and translate into hypertension later in life (fetal programming). Infants with low birthweight (LBW) are at increased risk for hypertension in addition to diabetes and obesity.¹² Mothers of LBW infants frequently have hypertension, obesity, preeclampsia, or malnutrition, and these maternal factors also carry an increased risk for hypertension in the infant. Brenner and colleagues¹³ proposed that this excess risk among LBW infants is driven by lower nephron number, which may predispose to the development of hypertension and the progressive loss of nephrons with aging. Indeed, maternal malnutrition in laboratory rats predisposes to small pups, low nephron number, and the future predisposition for hypertension. A study of nine white subjects with primary hypertension who died of accidental causes also found almost 50% fewer nephrons than in age- and sex-matched controls.¹⁴

Other studies, however, did not confirm a relationship between birthweight or low nephron number and hypertension. One study reported that LBW infants have a 25% risk for developing hypertension as an adult, but infants with high birthweight also carried a 20% risk.¹⁵

Thus, LBW and low nephron number likely reflect risk factors for development of hypertension rather than the underlying mechanism.

Role of the Immune System and Subtle Kidney Injury

Recently, a role of the immune system has been identified in primary hypertension.¹⁶ The initiating mechanism is not fully understood but may be initiated by renal vasoconstriction that results from a borderline hyperactive SNS, endothelial dysfunction, or oxidative stress and that is precipitated by diet (high salt or high sugar), metabolic mechanisms (hyperuricemia), emotional stress, a borderline hyperactive SNS, or other mechanisms.¹⁷

The renal vasoconstriction causes ischemia that stimulates low grade T-cell and macrophage infiltration (Fig. 35.4). Low-grade T-cell and macrophage infiltration in the interstitium occurs in 90% or more of kidney biopsy samples from patients with primary hypertension.¹⁸ Historically, the inflammatory and microvascular changes were considered the consequence of hypertensive kidney damage, but newer studies suggest that persistent kidney inflammation may have a role in causing the hypertensive response, possibly by causing renal vasoconstriction and impairing the ability to excrete sodium.

The mechanism is thought to be mediated by activation of the innate immune system followed by stimulation of adaptive immunity in which the T cells become sensitized to neoantigens, leading to an autoimmune mediated hypertension (Fig. 35.4). To date, two autoantigens have been identified, including heat shock protein 70 (HSP70) and oxidized isoketal-containing proteins (i.e., proteins oxidatively modified by highly reactive γ -ketoaldehydes).^{19,20} A role for HSP70 as a neoantigen is supported by experimental studies in which blocking T-cell sensitization to HSP70 can block hypertension, whereas sensitization of T cells to HSP70 in the kidney can induce a rise in BP. Clinical studies also have found evidence for both autoantibodies and T-cell sensitization to HSP70 in humans with primary hypertension.¹⁹ Isoketal accumulation in dendritic cells may also act as an autoantigen as it can induce costimulatory CD80 and CD86 proteins. Isoketal-modified proteins have been demonstrated in circulating dendritic cells and monocytes of hypertensive patients, and scavengers of isoketals prevent or ameliorate hypertension in several models of hypertension.²⁰

It is thought that the immune response to the autoantigens is orchestrated by T cells, B cells, and macrophages, with the CD8 T cells primarily driving the hypertensive response, whereas the regulatory T-cell populations counterregulate the response.^{16,21,22} Blocking the inflammation with immunosuppressive therapy can both prevent or correct the development of hypertension in experimental models.¹⁶ Pilot studies also suggest that blocking the immune system, such as by mycophenolate mofetil (MMF), can lower BP in subjects with primary hypertension.¹⁶

How Does Sodium Retention Lead to Hypertension?

Although the classical view has been that salt loading causes acute volume expansion followed by vasoconstriction and pressure-induced sodium excretion (pressure-natriuresis), in recent years the role of transient hypertonicity as driving the hypertensive response has become dominant (Fig. 35.5).

For example, an acute salt load increases blood pressure immediately through its effects to raise sodium concentration and vasopressin levels.²³ Hypertonicity also stimulates SNS activity in the brain that has systemic and kidney hemodynamic effects that increase BP and sodium retention. Hypertonicity also activates the immune response, stimulating dendritic cell activation, the production of interleukin-17 (IL-17) from T helper cells, cytokine release from T and B cells, and inducing immune cell infiltration in the kidney.^{24,25} Overproduction of IL-17, in addition to its effects on autoimmunity, also stimulates the

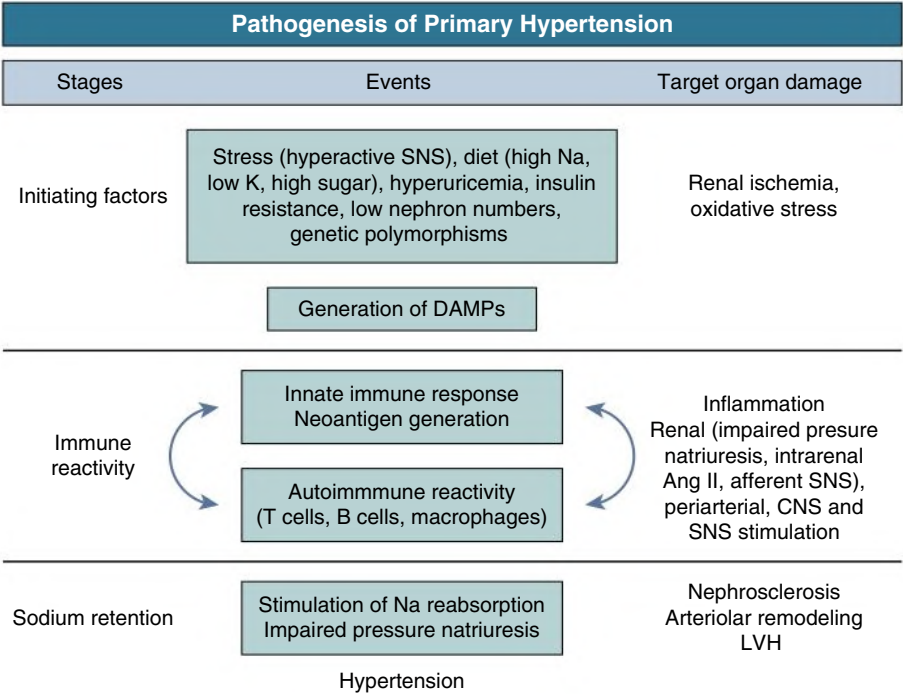


Fig. 35.4 Pathogenesis of Primary Hypertension. Initiating factors induce renal ischemia and oxidative stress with the stimulation of innate immune reactivity, neoantigen formation, and autoimmune reactivity that drives inflammation in the kidney, in the adventitia in the arteries, and in the central nervous system that stimulates the sympathetic nervous system. Sodium retention results from increased reabsorption and impaired pressure natriuresis as a consequence of renal inflammation. Established hypertension drives target organ damage. *Ang II*, angiotensin II; *CNS*, central nervous system; *DAMP*, damage-associated molecular patterns; *LVH*, left ventricular hypertrophy; *SNS*, sympathetic nervous system.

Hemodynamic Pathophysiology of Hypertension

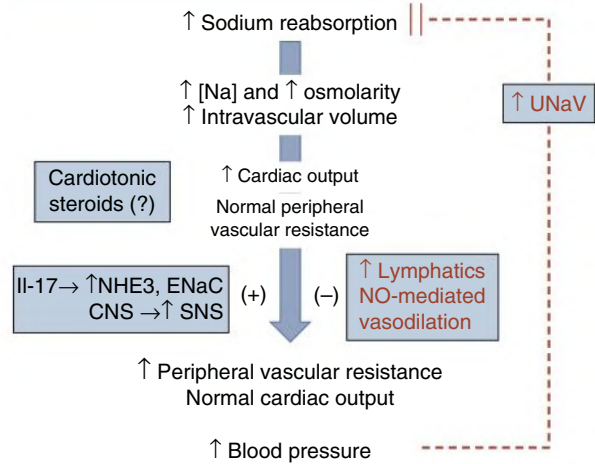


Fig. 35.5 Hemodynamic Pathophysiology of Hypertension. Sodium retention induces increase in sodium concentration, hyperosmolarity, and rise in intravascular volume with an increase in cardiac output, possibly driven in part by the generation of cardiotonic steroids. This transient state is followed by sustained increment in peripheral vascular resistance and normalization of the cardiac output. These events result from the combined effects of immune (IL-17) stimulated increase in sodium reabsorption and stimulation of the sympathetic nervous system (SNS). These effects are counterbalanced by vascular endothelial growth factor type C-mediated increase in the lymphatic network and endothelial nitric oxide (NO) generation induced by intravascular dilation (see text). The increase in blood pressure tends to correct sodium retention by stimulating pressure natriuresis. *CNS*, Central nervous system; *ENaC*, endothelial sodium channel; *NHE3*, sodium-hydrogen exchanger 3; *UNaV*, urinary sodium excretion.

sodium-hydrogen exchanger 3 (NHE3) in the proximal tubule and in the sodium chloride cotransporter (NCC) and epithelial sodium channel (ENaC) in the distal and collecting ducts, respectively, favoring sodium retention.²⁶

The hypertonicity and sodium retention also stimulates the release of cardiotonic steroids from the adrenal glands, such as ouabain and marinobufagenin, that partially block Na⁺,K⁺-ATPases, thereby stimulating natriuresis by downregulating kidney sodium transporters. However, they do this at the expense of raising blood pressure by increasing vasoconstriction and myocardial contractility by increasing cytosolic Ca²⁺ in vascular smooth muscle cells and cardiomyocytes.²⁷

There are also countering mechanisms that block the hypertensive response. Sodium retention activates the tonicity enhancer binding protein (TONEBP) in macrophages that produce vascular endothelial growth factor type C that stimulates the development of a lymphatic network that attenuates the hemodynamic effects of sodium retention.²⁸ Sodium retention may not be sequestered in an osmotically independent compartment²⁹ as originally considered²⁸; nevertheless, the role of macrophages is well demonstrated because macrophage depletion accentuates hypertension. Another mechanism that counters the salt-induced increase in blood pressure is the endothelial production of nitric oxide stimulated by intravascular volume expansion.³⁰

Pathogenic Mechanisms Driving the Current Epidemic of Hypertension

The rise of hypertension in the last century is largely thought to be the consequence of the rise in obesity, and also the increase in life span that occurred during that time. Although obesity-associated hypertension likely involves all three mechanisms previously discussed, it may also result from other related mechanisms, including diet (salt and sugar), subtle kidney injury, coexistence of endothelial



Fig. 35.6 Grades of Hypertensive Retinopathy. (A) Mild hypertensive retinopathy with arteriolar narrowing and arteriovenous nicking. (B) Moderate hypertensive retinopathy with cotton-wool spots (nerve fiber layer infarcts) and arteriovenous nicking. (C) Papilledema, cotton-wool spots, macular yellow exudates (star formation pattern), and retinal hemorrhages in a patient with hypertensive emergency. (Courtesy J. Kinyoun, University of Washington.)

dysfunction and oxidative stress, activation of the SNS and intrarenal renin-angiotensin-aldosterone system (RAAS), and effects of hyperleptinemia and hyperinsulinemia.^{31,32} Chronically elevated leptin levels, which are common in obese persons, can activate the SNS in the central nervous system (CNS) through proopiomelanocortin neurons that activate melanocortin-4 receptors.³²

Other proposed mechanisms include hyperuricemia, possibly driven by intake of fructose-containing sugars and purine-rich foods.¹⁷ Studies also have implicated a hyperactive SNS in early hypertension, particularly in young or borderline hypertensive patients.³³ Postulated mechanisms include a defect in baroreceptor sensitivity and an increase in SNS response to emotional or work-related stress. Activation of either the systemic or the local renin-angiotensin system (RAS) is also common in hypertension. Whereas plasma renin activity is elevated in 20% of patients, renin activity is either normal (50%) or low (30%) in the majority. However, normal renin activity may be inappropriately high in relation to the total body sodium.³⁴ Some hypertensive patients also have elevated plasma aldosterone, especially if the RAS is inhibited with angiotensin-converting enzyme (ACE) inhibitors or angiotensin receptor blockers (ARBs).³⁵ This condition is known as *aldosterone breakthrough* and is often observed in persons who are obese or have hyperinsulinemia.

CLINICAL MANIFESTATIONS

Evaluation of a patient with hypertension requires careful history and physical examination, an evaluation of risk factors for hypertension, a search for potential secondary causes, and evaluation for end organ damage.

Blood pressure should be measured on at least three occasions to confirm persistent hypertension using the techniques described in [Chapter 34](#). Home BP monitoring or 24-hour ambulatory BP monitoring is recommended to determine whether the hypertension is white coat hypertension or masked hypertension. White coat and masked hypertension can be associated with end organ disease, including left ventricular hypertrophy (LVH) and mildly increased albuminuria; diagnosis should be followed by assessment of CV risk factors and frequent reevaluation of BP.

The history should investigate the onset and duration of hypertension and the presence of a family history of hypertension or cardiorenal disease. The history should identify risk factors for hypertension (obesity, diabetes, physical activity, gout, alcohol, smoking, diet, emotional or work-related stress) and any hypertension-related morbidity

as well as consumption of over-the-counter and prescribed medications that may increase BP.

Hypertension is often asymptomatic, but even childhood-associated hypertension can be associated with impaired memory and mental performance, and hypertension remains a major risk factor for vascular dementia. Hypertension, especially when severe (BP >160/100 mm Hg), also may be associated with headache, classically occipital and pulsatile. In hypertensive emergency, encephalopathy rarely occurs, sometimes with altered mental status and seizures. Rarely, patients may lose vision from papilledema. Hypertension, especially over 160/100 mm Hg, also places individuals at acute risk for myocardial infarction (MI), congestive heart failure (CHF) with pulmonary edema, aortic dissection, cerebrovascular accident (stroke), and kidney failure.

Physical examination includes BP measurement in both arms and cardiac examination. Attention should be focused on both the large vessels (by both palpation and listening for bruits) and the retina to evaluate the microvasculature ([Fig. 35.6](#)). Lab tests should include hematocrit, electrolytes, creatinine (and estimated GFR), calcium and phosphate (to look for primary hyperparathyroidism), fasting lipid profile (cholesterol and triglycerides), uric acid, and urinalysis. A chest radiograph and electrocardiogram should be performed to assess cardiac size and look for aortic dilation.

Additional tests include 24-hour urine sodium and potassium excretion. Urinary Na⁺ and K⁺ excretion correlates with intake if the patient is in steady state (desirable values are <100 mmol/L Na⁺ and >100 mmol/L K⁺ in 24 hours). A spot urine albumin-creatinine and an echocardiogram may uncover additional evidence of end organ damage, especially in diabetic patients and those with cardiac problems, respectively ([Fig. 35.7](#)).

PATHOLOGY

Primary hypertension is characterized by arteriosclerosis of the preglomerular afferent arteriole and interlobular artery. Arteriosclerosis, seen in 90% of patients, involves the replacement of smooth muscle cells of the media in the afferent arteriole with connective tissue, often with hyaline material (plasma proteins) in the subintima (hyalinosis)¹⁸ ([Fig. 35.8](#)). Glomerular and tubular ischemia are common, with shrinkage of the glomerular tuft, tubular atrophy, and interstitial fibrosis. Occasionally, glomerulosclerosis and severe tubulointerstitial injury are seen. With hypertensive emergency, a proliferative arteriolopathy occurs, occasionally with fibrinoid necrosis. Concentric layers of connective tissue and cells may give an onion-skin appearance to

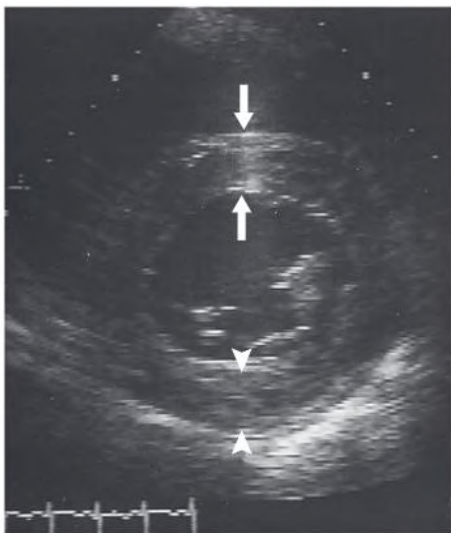


Fig. 35.7 Echocardiogram Showing Concentric Left Ventricular Hypertrophy. Septal thickness (between arrows) and posterior wall thickness (between arrowheads) are increased (to 16 mm) in a patient with primary hypertension (normal is ≤ 11 mm). (Courtesy A. Pearlman, University of Washington.)

the intima, which may progress to a total obliteration of the lumen. In addition to vascular changes, immune cell infiltration in tubulointerstitial regions is present even at stages when the vascular changes are minimal or nonexistent.^{18,36}

DIAGNOSIS

Diagnosis of primary hypertension requires the elimination of secondary causes. Suspicion of secondary causes of hypertension should be raised by finding new onset hypertension in very young or elderly patients (renovascular hypertension), drug-resistant hypertension, hypokalemia in the absence of being on diuretic therapy (primary aldosteronism from hyperplasia or tumors), or with impaired kidney function (kidney parenchymal disease). Medications that increase BP include nonsteroidal antiinflammatory drugs, corticosteroids, sympathomimetics, and oral contraceptives, as well as excessive alcohol intake and cocaine use. Table 34.9 in Chapter 34 provides a list of secondary causes along with recommended evaluations.

NATURAL HISTORY

Although hypertension is often asymptomatic (referred to as the “silent killer”), it represents a major risk factor for CV mortality, especially related to pressure-related morbidities such as heart failure (HF), stroke, aortic aneurysms and dissection, chronic kidney disease (CKD), and vascular dementia. Hypertension is also a risk factor for coronary artery disease, although the association is less strong.

Increased Risk for Cardiovascular Disease

Hypertension is the most common cause of both stroke and HF associated with preserved ejection fraction (Fig. 35.9).³⁷ In essence, hypertension causes concentric LVH with supernormal systolic function. However, subendocardial inflammation and fibrosis also occur, resulting in a stiffer myocardium and impairing diastolic filling and eventually causing heart failure. Almost 80% to 90% of patients with heart failure have a history of hypertension.³⁸

Hypertension can also cause vascular changes that can lead to aortic dissection, cerebral and aortic aneurysms, and stiffening of blood vessels from changes in collagen content that can result in higher pulse pressures.^{39,40} Both elevated systolic BP and elevated pulse pressure confer increased risks for stroke, but high pulse pressure is additionally complicated as it may limit the magnitude with which systolic BP can be lowered.⁴¹ Studies in people with diabetes suggested that diastolic BP should not be lowered below 60 mm Hg as CV risk will start to increase.⁴² A post hoc analysis of the ONTARGET and TRANSCEND trials support the notion that DBP less than 70 mm Hg or greater than 80 mm Hg is associated with a higher risk in patients with an SBP of 120 to 140 mm Hg. These findings support European guidelines, which take DBP at optimal SBP control into consideration.¹

Hypertension also increases the risk of coronary heart disease (Fig. 35.9), peripheral vascular disease, and carotid atherosclerosis with or without cerebral emboli. Hypertension is also associated with increased risk for vascular dementia.

Increased Risk for Chronic Kidney Disease

Most patients with newly diagnosed primary hypertension have normal kidney function, stage 1 CKD (GFR ≥ 90 mL/min/1.73 m²), or stage 2 CKD (GFR 60–90 mL/min/1.73 m²) with or without mildly increased albuminuria (formerly termed microalbuminuria). Despite relatively preserved kidney function, kidney biopsy, if done, usually shows early arteriolosclerosis, hyalinosis, and interstitial inflammation (see Fig. 35.8).

Mildly increased albuminuria, which is a marker of vascular disease and CV risk, occurs in 15% to 30% of patients, whereas more severe albuminuria is uncommon and nephrotic-range proteinuria rare.⁴³ Elevations in serum creatinine over time develop in 10% to 20% of patients with poorly controlled BP, and the risk is greater in diabetic patients, African Americans, the elderly, patients with gout or hyperuricemia, and those with higher SBP (>160 mm Hg).⁴⁴ In 2% to 5% of those with poorly controlled SBP (>160 mm Hg), progression to kidney failure will occur over 10 to 15 years (Figs. 35.10 and 35.11).^{45,46} Despite the relative infrequency for primary hypertension to progress to kidney failure, hypertension is recorded as the second most common cause of treated kidney failure after diabetes in the United States and Europe.⁴⁷ Furthermore, almost all patients with diabetes have hypertension when they start dialysis.

The incidence of kidney failure in African Americans with hypertension is twofold to sixfold greater than in Whites (Fig. 35.11).^{46,47} Kidney biopsies of African Americans with hypertension show more severe hypertensive injury, with more prominent vascular changes and increased frequency of segmental and global glomerulosclerosis. The increased frequency of glomerulosclerosis may be caused in part by a polymorphism in apolipoprotein L1 (APOL1), the product of which is expressed in the podocyte.⁴⁸ The mechanism by which APOL1 polymorphisms increase the risk for progression of kidney disease is unclear but may relate to effects on the podocyte that include lysosomal swelling, mitochondrial dysfunction, and blockade of autophagy.^{49,49a}

Although hypertension is a risk factor for CKD, as CKD progresses the prevalence and severity of hypertension also increases and requires more medications to achieve control.⁵⁰ Multiple mechanisms likely account for this, including sodium retention, increased afferent sympathetic tone induced by kidney inflammation,⁵¹ activation of the intrarenal RAS,⁵² endothelial dysfunction,⁵³ and the frequent occurrence of sleep apnea. These effects are associated with loss of nocturnal BP dipping, and BP may even increase at night, which is associated with an increase in CV complications and progression of CKD.⁵⁴

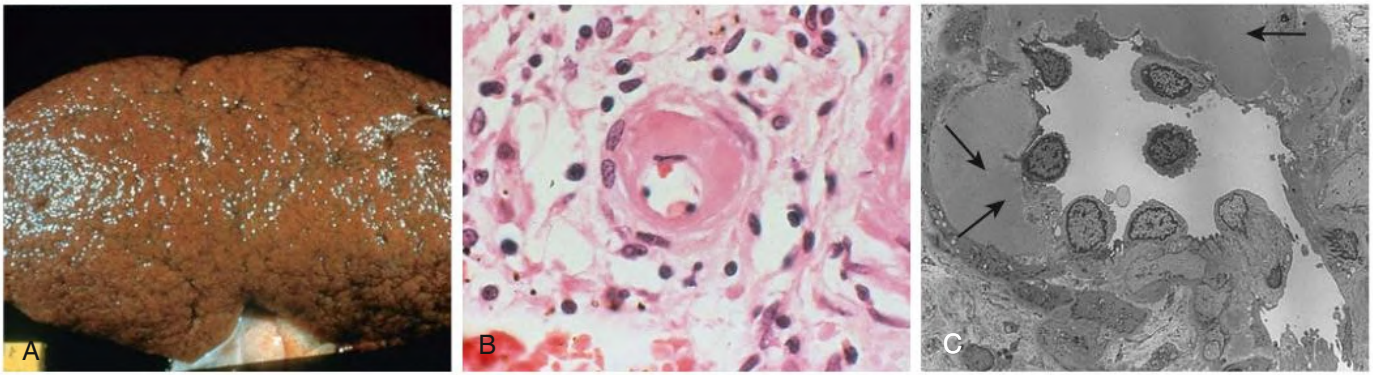
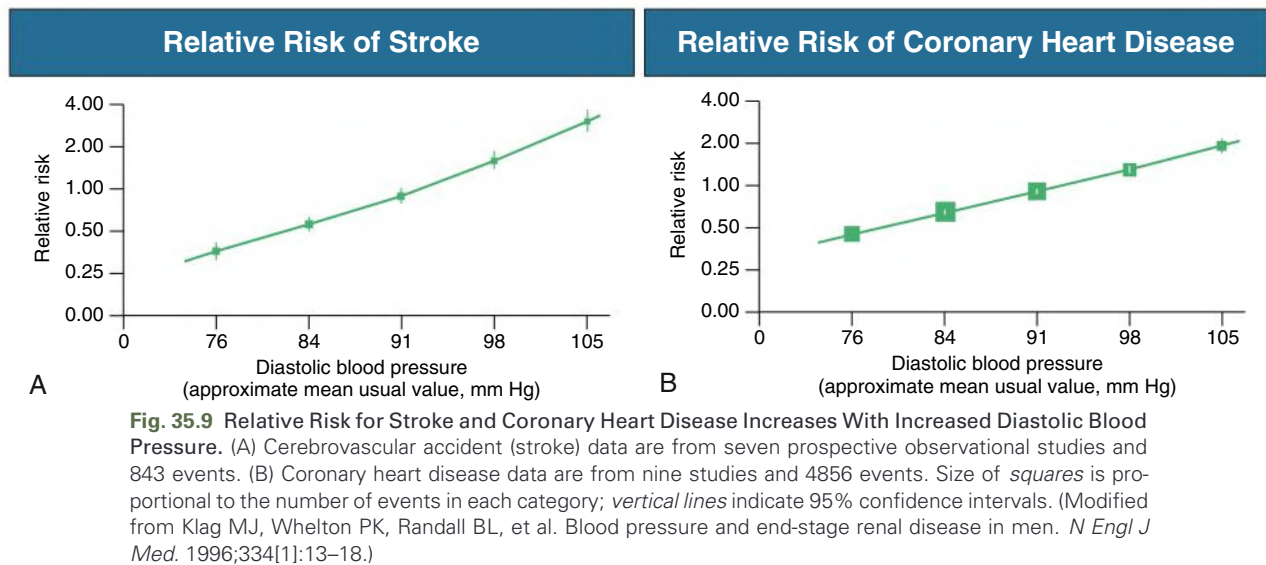


Fig. 35.8 Kidney Pathology in Primary Hypertension. (A) A granular pitted kidney in a patient with chronic primary hypertension. (B) Arteriolosclerosis with subintimal hyalinosis. (C) Electron micrograph showing hyalinosis with the accumulation of insudative plasma proteins (arrows) in the subendothelium of an arteriole. (A, Courtesy Harvard Medical School. B–C, Courtesy C.E. Alpers, University of Washington.)



End-Stage Kidney Disease and Blood Pressure

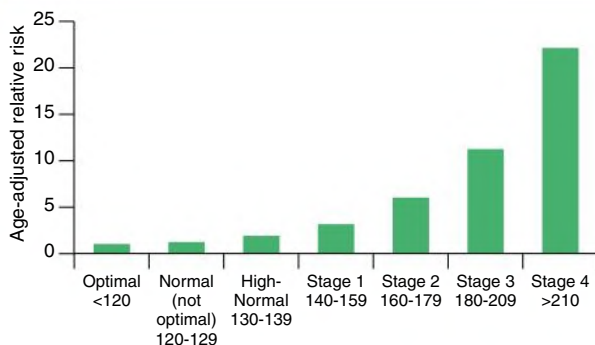


Fig. 35.10 End-Stage Kidney Disease and Blood Pressure. Incidence of treated kidney failure related to baseline blood pressure in the MRFIT study. Blood pressure stages were based on definitions at that time. Mean follow-up was 16 years. (From Perry HM Jr, Miller JP, Fornoff JR, et al. Early predictors of 15-year end-stage renal disease in hypertensive patients. *Hypertension*. 1995;25[4 Pt 1]:587–594.)

Effect of Race on Incidence of End-Stage Kidney Disease in Hypertension

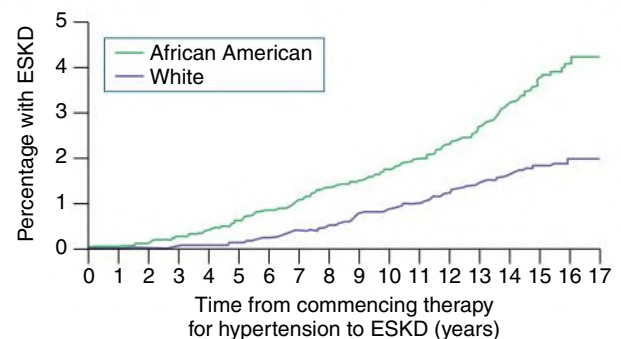


Fig. 35.11 Effect of Race on Incidence of Treated Kidney Failure Disease in Hypertensive Patients. The cumulative incidence of treated kidney failure (end-stage kidney disease (ESKD) in African-American and White hypertensive veterans (Kaplan-Meier estimates). (From Seravalle G, Grassi G. Obesity and hypertension. *Pharmacol Res*. 2017;122:1–7.)

TABLE 35.3 Meta-analysis of Effect of Antihypertensive Agents on Cardiovascular Outcomes in Hypertensive Patients

Outcome	Number of Trials	Effects Model	RR (95% CI)	P Value for Heterogeneity
Coronary heart disease	24	Fixed	0.86 (0.80–0.93)	.55
		Random	0.87 (0.80–0.94)	.55
Stroke	23	Fixed	0.69 (0.64–0.74)	.004
		Random	0.68 (0.61–0.76)	.004
CHF	7	Fixed	0.54 (0.45–0.66)	.66
		Random	0.60 (0.49–0.74)	.80
Major CVD events	28	Fixed	0.78 (0.74–0.81)	<.001
		Random	0.73 (0.62–0.87)	<.001
CVD mortality	23	Fixed	0.84 (0.78–0.90)	.10
		Random	0.84 (0.78–0.90)	.10
Total mortality	25	Fixed	0.90 (0.85–0.95)	.58
		Random	0.90 (0.85–0.95)	.59

Analysis was based on 42 clinical trials that included 192,478 patients randomized to seven major treatment strategies, including placebo.

CHF, Congestive heart failure; CI, confidence interval; CVD, cardiovascular disease; RR, relative risk.

From Psaty BM, Lumley T, Furberg CD, et al. Health outcomes associated with various antihypertensive therapies used as first-line agents: a network meta-analysis. *JAMA*. 2003;289:2534–2544.

Effects of Antihypertensive Therapy on Cardiovascular Disease and Kidney Outcomes

Effect of BP Control on Cardiovascular Mortality

Blood pressure control has been consistently shown to reduce CV outcomes (especially stroke and heart failure) and vascular dementia (Table 35.3).^{55,56} The Systolic Blood Pressure Intervention Trial (SPRINT) found that lowering SBP to 120 to 130 mm Hg provided greater CV protection in nondiabetic subjects than a target BP of 140 mm Hg.⁵⁷ However, intensive BP lowering treatment carried a higher risk of hypotension, syncope, and acute hemodynamically associated kidney injury, which resolved within a month in almost 90% of the cases without change in BP goal.⁵⁸ However, the SPRINT study showed that the benefits of lowering BP outweigh the associated risks. In contrast, a similar study, known as the Action to Control Cardiovascular Risk in Diabetes Blood Pressure (ACCORD), did not demonstrate overall CV protection in diabetics in which SBP was targeted to 120 mm Hg compared with 140 mm Hg.¹⁴ Nevertheless, recent analyses suggest a CV risk reduction, with more intensive BP treatment with achieved BP closer to 125 mm Hg.^{59,60}

Although lowering BP protects against progression of CKD, the target BP has been controversial. Intensive BP control in the SPRINT trial did not provide any additional protection for CKD progression, but the SPRINT trial did not enroll patients with more than 1 g proteinuria per day. However, the Modification of Diet in Renal Disease (MDRD) and the African American Study of Kidney Disease (AASKD) studies did enroll patients with significant albuminuria and found less kidney disease progression in this subgroup with intensive BP lowering.^{61,62} Despite the lack of benefit on kidney disease in patients without severe albuminuria (<300 mg/day), the benefit on CV protection suggests that lower BP targets should be sought in these patients. These findings have led to the general recommendation of a target BP of 130/80 mm Hg for all patients, including the elderly, those with diabetes, and those with CKD by the American College of Cardiology, the European Guidelines,¹ and by the American Diabetes Association.⁶³

A discussion of pharmacologic management of hypertension is provided in Chapter 37. Blood pressure control in patients with high CV risk and/or CKD usually requires at least two medications, of which one should be a RAAS blocker at maximally tolerated doses. In the AASKD study, use of an ACE inhibitor (ramipril) was more

Ramipril Is Superior to Amlodipine in Reducing Renal Events in Hypertensive African Americans with Renal Impairment

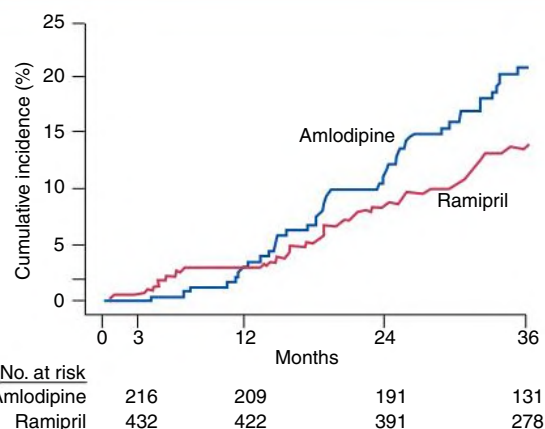


Fig. 35.12 Ramipril is Superior to Amlodipine in Reducing Kidney Events in Hypertensive African Americans With Mild to Moderate Kidney Impairment. The angiotensin-converting enzyme inhibitor ramipril resulted in fewer kidney end points (proteinuria, decline in kidney function, treated kidney failure, death) compared with the dihydropyridine calcium channel blocker amlodipine, in the African American Study of Kidney Disease. (From Agodoa LY, Appel L, Bakris GL, et al. Effect of ramipril vs amlodipine on renal outcomes in hypertensive nephrosclerosis: a randomized controlled trial. *JAMA*. 2001;285[21]:2719–2728.)

effective at slowing CKD progression than either amlodipine (Fig. 35.12) or metoprolol.⁶⁴ Similar results were seen in the Irbesartan Diabetic Nephropathy Trial (IDNT) trial, where irbesartan was superior to amlodipine in slowing diabetic nephropathy progression.⁶⁵ Furthermore, in the high-risk hypertensive population, there is also evidence from the ACCOMPLISH trial that better CV protection can be provided if the ACE inhibitor is combined with a calcium channel blocker rather than a thiazide diuretic.⁶⁶

In people with CKD and hypertension, a diuretic is often required. Nevertheless, diuretics should be used only as needed. Some studies

suggest that thiazide and thiazide-like diuretics are associated with worsening of kidney function in patients with hypertension. In the European Working Party on High Blood Pressure in the Elderly trial⁶⁷ and in the Systolic Hypertension in the Elderly trial,⁶⁸ serum creatinine increased significantly in patients treated with thiazide diuretics compared with placebo. Similarly, in the Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial (ALLHAT), the chlorthalidone-treated group showed statistically worse kidney function than either the amlodipine or the lisinopril group at both 2- and 4-year end points despite its better reduction of strokes and heart failure.⁶⁹ This could likely be accounted for by volume depletion in many cases, but diuretics have been shown to induce mild kidney injury in various animal models, possibly because of hypokalemia, hyperuricemia, and stimulation of the RAAS.⁷⁰

Acute BP lowering with any medication may cause a rise in serum creatinine of up to 30% in the first month of therapy, and treatment should be continued if there are no electrolyte abnormalities or hypotension.^{71,72} In numerous trials including SPRINT and ACCORD, kidney disease progression is slowed and CV risk (especially from heart failure) markedly reduced. Furthermore, RAAS blockade should not be stopped due to small, limited increases in creatinine, as a retrospective analysis found that the discontinuation of ACE inhibitor or ARB therapy for acute decreases in eGFR (<30 mL/min/1.73 m²) was not associated with increased risk of kidney failure but carried a 39% higher risk of mortality and 37% increase in major adverse CV events (MACE).⁷³

SELF-ASSESSMENT QUESTIONS

1. Primary hypertension is characterized by all of the following *except*:
 - A. Primary hypertension is more common in older patients.
 - B. Salt sensitivity is less common in older patients.
 - C. Primary hypertension is associated with a reduction in kidney blood flow and usually mild reduction in GFR.
 - D. Primary hypertension is more common in individuals with low birthweight and low nephron number.
 - E. The presence of the *APOL1* polymorphism is associated with more rapid progression of kidney disease in African Americans with hypertension.
2. The kidney pathology in patients with primary hypertension may consist of all of the following *except*:
 - A. arteriolosclerosis with arteriolar fibrosis or hyalinosis.
 - B. evidence of tubular ischemia change is commonly present.
 - C. glomerulosclerosis may be present, especially in African Americans.
 - D. kidney histologic studies can rarely appear normal.
 - E. transmural necrosis of medium-sized vessels can occasionally be observed.
3. Which of the following is *not* a dietary risk factor for primary hypertension?
 - A. Sodium
 - B. Potassium
 - C. Sugar
 - D. Alcohol
 - E. Coffee

REFERENCES

- Williams B, Mancia G, Spiering W, et al. 2018 ESC/ESH Guidelines for the management of arterial hypertension. *Eur Heart J*. 2018;39(33):3021–3104.
- Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA guideline for the prevention, detection, evaluation, and management of high blood pressure in adults: executive summary: a report of the American College of Cardiology/American Heart Association task force on clinical practice guidelines. *J Am Coll Cardiol*. 2018;71(19):2199–2269.
- Bevan AT, Honour AJ, Stott FH. Direct arterial pressure recording in unrestricted man. *Br Heart J*. 1969;31(3):387–388.
- Mills KT, Stefanescu A, He J. The global epidemiology of hypertension. *Nat Rev Nephrol*. 2020;16(4):223–237.
- Burt VL, Whelton P, Roccella EJ, et al. Prevalence of hypertension in the US adult population. Results from the third national health and nutrition examination survey, 1988–1991. *Hypertension*. 1995;25(3):305–313.
- Rodriguez-Iturbe B, Johnson RJ. Genetic polymorphisms in hypertension: are we missing the immune connection? *Am J Hypertens*. 2019;32(2):113–122.
- Lifton RP. Genetic dissection of human blood pressure variation: common pathways from rare phenotypes. *Harvey Lect*. 2004;100:71–101.
- Guyton AC, Coleman TG, Cowley Jr AV, Scheel KW, Manning Jr RD, Norman Jr RA. Arterial pressure regulation. overriding dominance of the kidneys in long-term regulation and in hypertension. *Am J Med*. 1972;52(5):584–594.
- He FJ, Tan M, Ma Y, MacGregor GA. Salt reduction to prevent hypertension and cardiovascular disease: JACC state-of-the-art review. *J Am Coll Cardiol*. 2020;75(6):632–647.
- Ehret GB, Munroe PB, Rice KM, et al. Genetic variants in novel pathways influence blood pressure and cardiovascular disease risk. *Nature*. 2011;478(7367):103–109.
- Padmanabhan S, Caulfield M, Dominiczak AF. Genetic and molecular aspects of hypertension. *Circ Res*. 2015;116(6):937–959.
- Kanda T, Murai-Takeda A, Kawabe H, Itoh H. Low birth weight trends: possible impacts on the prevalences of hypertension and chronic kidney disease. *Hypertens Res*. 2020;43(9):859–868.
- Brenner BM, Garcia DL, Anderson S. Glomeruli and blood pressure. Less of one, more the other? *Am J Hypertens*. 1988;1(4 Pt 1):335–347.
- Keller G, Zimmer G, Mall G, Ritz E, Amann K. Nephron number in patients with primary hypertension. *N Engl J Med*. 2003;348(2):101–108.
- Eriksson J, Forsen T, Tuomilehto J, Osmond C, Barker D. Fetal and childhood growth and hypertension in adult life. *Hypertension*. 2000;36(5):790–794.
- Rodriguez-Iturbe B, Pons H, Johnson RJ. Role of the immune system in hypertension. *Physiol Rev*. 2017;97(3):1127–1164.
- Sanchez-Lozada LG, Rodriguez-Iturbe B, Kelley EE, et al. Uric acid and hypertension: an update with recommendations. *Am J Hypertens*. 2020;33(7):583–594.
- Sommers SC, Relman AS, Smithwick RH. Histologic studies of kidney biopsy specimens from patients with hypertension. *Am J Pathol*. 1958;34(4):685–715.
- Pons H, Ferrebuz A, Quiroz Y, et al. Immune reactivity to heat shock protein 70 expressed in the kidney is cause of salt-sensitive hypertension. *Am J Physiol Renal Physiol*. 2013;304(3):F289–F299.
- Kirabo A, Fontana V, de Faria AP, et al. DC isoketal-modified proteins activate T cells and promote hypertension. *J Clin Invest*. 2014;124(10):4642–4656.
- Guzik TJ, Hoch NE, Brown KA, et al. Role of the T cell in the genesis of angiotensin II induced hypertension and vascular dysfunction. *J Exp Med*. 2007;204(10):2449–2460.
- Chan CT, Sobey CG, Lieu M, et al. Obligatory role for B cells in the development of angiotensin II-dependent hypertension. *Hypertension*. 2015;66(5):1023–1033.
- Kanbay M, Aslan G, Afsar B, et al. Acute effects of salt on blood pressure are mediated by serum osmolality. *J Clin Hypertens*. 2018;20(10):1447–1454.
- Wu C, Yosef N, Thalhammer T, et al. Induction of pathogenic TH17 cells by inducible salt-sensing kinase SGK1. *Nature*. 2013;496(7446):513–517.
- Kleinewietfeld M, Manzel A, Titze J, et al. Sodium chloride drives autoimmune disease by the induction of pathogenic TH17 cells. *Nature*. 2013;496(7446):518–522.
- Norlander AE, Saleh MA, Kamat NV, et al. Interleukin-17A regulates renal sodium transporters and renal injury in angiotensin II-induced hypertension. *Hypertension*. 2016;68(1):167–174.
- Bagrov AY, Shapiro JL, Fedorova OV. Endogenous cardiotonic steroids: physiology, pharmacology, and novel therapeutic targets. *Pharmacol Rev*. 2009;61(1):9–38.
- Titze J, Luft FC. Speculations on salt and the genesis of arterial hypertension. *Kidney Int*. 2017;91(6):1324–1335.
- Rossitto G, Mary S, Chen JY, et al. Tissue sodium excess is not hypertonic and reflects extracellular volume expansion. *Nat Commun*. 2020;11(1):4222.
- Morris Jr RC, Schmidlin O, Sebastian A, Tanaka M, Kurtz TW. Vasodysfunction that involves renal vasodysfunction, not abnormally increased renal retention of sodium, accounts for the initiation of salt-induced hypertension. *Circulation*. 2016;133(9):881–893.
- Seravalle G, Grassi G. Obesity and hypertension. *Pharmacol Res*. 2017;122:1–7.
- Hall JE, da Silva AA, do Carmo JM, et al. Obesity-induced hypertension: role of sympathetic nervous system, leptin, and melanocortins. *J Biol Chem*. 2010;285(23):17271–17276.
- Grassi G, Ram VS. Evidence for a critical role of the sympathetic nervous system in hypertension. *J Am Soc Hypertens*. 2016;10(5):457–466.
- Brunner HR, Laragh JH, Baer L, et al. Essential hypertension: renin and aldosterone, heart attack and stroke. *N Engl J Med*. 1972;286(9):441–449.
- Bombard AS, Rekhman Y, Klemmer PJ, Canetta PA, Radhakrishnan J, Appel GB. Aldosterone breakthrough during aliskiren, valsartan, and combination (aliskiren + valsartan) therapy. *J Am Soc Hypertens*. 2012;6(5):338–345.
- Hughson MD, Gobe GC, Hoy WE, Manning Jr RD, Douglas-Denton R, Bertram JF. Associations of glomerular number and birth weight with clinicopathological features of African Americans and whites. *Am J Kidney Dis*. 2008;52(1):18–28.
- MacMahon S, Peto R, Cutler J, et al. Blood pressure, stroke, and coronary heart disease. part 1, prolonged differences in blood pressure: prospective observational studies corrected for the regression dilution bias. *Lancet*. 1990;335(8692):765–774.
- Yancy CW, Jessup M, Bozkurt B, et al. 2016 ACC/AHA/HFSA focused update on new pharmacological therapy for heart failure: an update of the 2013 ACCF/AHA guideline for the management of heart failure: a report of the American College of Cardiology/American Heart Association task force on clinical practice guidelines and the Heart Failure Society of America. *J Am Coll Cardiol*. 2016;68(13):1476–1488.
- Nagao K, Inada T, Tamura A, et al. Circulating markers of collagen types I, III, and IV in patients with dilated cardiomyopathy: relationships with myocardial collagen expression. *ESC Heart Fail*. 2018;5(6):1044–1051.
- Hung CS, Chou CH, Liao CW, et al. Aldosterone induces tissue inhibitor of metalloproteinases-1 expression and further contributes to collagen accumulation: from clinical to bench studies. *Hypertension*. 2016;67(6):1309–1320.
- Bohm M, Schumacher H, Teo KK, et al. Achieved diastolic blood pressure and pulse pressure at target systolic blood pressure (120–140 mmHg) and cardiovascular outcomes in high-risk patients: results from ontarget and transcend trials. *Eur Heart J*. 2018;39(33):3105–3114.
- Gomadam P, Shah A, Qureshi W, et al. Blood pressure indices and cardiovascular disease mortality in persons with or without diabetes mellitus. *J Hypertens*. 2018;36(1):85–92.
- Bakris GL, Molitch M. Microalbuminuria as a risk predictor in diabetes: the continuing saga. *Diabetes Care*. 2014;37(3):867–875.
- Johnson RJ, Segal MS, Srinivas T, et al. Essential hypertension, progressive renal disease, and uric acid: a pathogenetic link? *J Am Soc Nephrol*. 2005;16(7):1909–1919.
- Klag MJ, Whelton PK, Randall BL, et al. Blood pressure and end-stage renal disease in men. *N Engl J Med*. 1996;334(1):13–18.

46. Perry Jr HM, Miller JP, Fornoff JR, et al. Early predictors of 15-year end-stage renal disease in hypertensive patients. *Hypertension*. 1995;25(4 Pt 1):587–594.
47. Sarafidis PA, Li S, Chen SC, et al. Hypertension awareness, treatment, and control in chronic kidney disease. *Am J Med*. 2008;121(4):332–340.
48. Lipkowitz MS, Freedman BI, Langefeld CD, et al. Apolipoprotein L1 gene variants associate with hypertension-attributed nephropathy and the rate of kidney function decline in African Americans. *Kidney Int*. 2013;83(1):114–120.
49. Kumar V, Singhal PC. APOL1 and kidney cell function. *Am J Physiol Renal Physiol*. 2019;317(2):F463–F477.
- 49a. Wu J, Raman A, Coffey NJ, et al. The key role of NLRP3 and STING in APOL1-associated podocytopathy. *J Clin Invest*. 2021;131(20):e136329.
50. Muntner P, Anderson A, Charleston J, et al. Hypertension awareness, treatment, and control in adults with CKD: results from the Chronic Renal Insufficiency Cohort (CRIC) study. *Am J Kidney Dis*. 2010;55(3):441–451.
51. Converse Jr RL, Jacobsen TN, Toto RD, et al. Sympathetic overactivity in patients with chronic renal failure. *N Engl J Med*. 1992;327(27):1912–1918.
52. Greene EL, Kren S, Hostetter TH. Role of aldosterone in the remnant kidney model in the rat. *J Clin Invest*. 1996;98(4):1063–1068.
53. Baylis C. Nitric oxide deficiency in chronic kidney disease. *Am J Physiol Renal Physiol*. 2008;294(1):F1–F9.
54. Mojon A, Ayala DE, Pineiro L, et al. Comparison of ambulatory blood pressure parameters of hypertensive patients with and without chronic kidney disease. *Chronobiol Int*. 2013;30(1–2):145–158.
55. Psaty BM, Lumley T, Furberg CD, et al. Health outcomes associated with various antihypertensive therapies used as first-line agents: a network meta-analysis. *JAMA*. 2003;289(19):2534–2544.
56. Forette F, Seux ML, Staessen JA, et al. Prevention of dementia in randomised double-blind placebo-controlled Systolic hypertension in Europe (Syst-Eur) trial. *Lancet*. 1998;352(9137):1347–1351.
57. Wright Jr JT, Whelton PK, Reboussin DM. A randomized trial of intensive versus standard blood-pressure control. *N Engl J Med*. 2016;374(23):2294.
58. Malhotra R, Craven T, Ambrosius WT, et al. Effects of intensive blood pressure lowering on kidney tubule injury in CKD: a longitudinal subgroup Analysis in SPRINT. *Am J Kidney Dis*. 2019;73(1):21–30.
59. Buckley LF, Dixon DL, Gft W, Wijesinghe DS, Baker WL, Van Tassel BW. Effect of intensive blood pressure control in patients with type 2 diabetes mellitus over 9 years of follow-up: a subgroup analysis of high-risk accord trial participants. *Diabetes Obes Metab*. 2018;20(6):1499–1502.
60. Collard D, Brouwer TF, Peters RJG, Vogt L, van den Born BH. Creatinine rise during blood pressure therapy and the risk of adverse clinical outcomes in patients with type 2 diabetes mellitus. *Hypertension*. 2018;72(6):1337–1344.
61. Appel LJ, Wright Jr JT, Greene T, et al. Intensive blood-pressure control in hypertensive chronic kidney disease. *N Engl J Med*. 2010;363(10):918–929.
62. Sarnak MJ, Greene T, Wang X, et al. The effect of a lower target blood pressure on the progression of kidney disease: long-term follow-up of the modification of diet in renal disease study. *Ann Intern Med*. 2005;142(5):342–351.
63. Association AD. Standards of medical care in diabetes-2021. *Diabetes Care*. 2021 (in press).
64. Agodoa LY, Appel L, Bakris GL, et al. Effect of ramipril vs amlodipine on renal outcomes in hypertensive nephrosclerosis: a randomized controlled trial. *JAMA*. 2001;285(21):2719–2728.
65. Lewis EJ, Hunsicker LG, Clarke WR, et al. Renoprotective effect of the angiotensin-receptor antagonist irbesartan in patients with nephropathy due to type 2 diabetes. *N Engl J Med*. 2001;345(12):851–860.
66. Jamerson K, Weber MA, Bakris GL, et al. Benazepril plus amlodipine or hydrochlorothiazide for hypertension in high-risk patients. *N Engl J Med*. 2008;359(23):2417–2428.
67. Fletcher A, Amery A, Birkenhager W, et al. Risks and benefits in the trial of the European working party on high blood pressure in the elderly. *J Hypertens*. 1991;9(3):225–230.
68. Savage PJ, Pressel SL, Curb JD, et al. Influence of long-term, low-dose, diuretic-based, antihypertensive therapy on glucose, lipid, uric acid, and potassium levels in older men and women with isolated systolic hypertension: the systolic hypertension in the elderly program. SHEP Cooperative Research Group. *Arch Intern Med*. 1998;158(7):741–751.
69. Major outcomes in high-risk hypertensive patients randomized to angiotensin-converting enzyme inhibitor or calcium channel blocker vs diuretic: the Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial (ALLHAT). *JAMA*. 2002;288(23):2981–2997.
70. Reungjui S, Hu H, Mu W, et al. Thiazide-induced subtle renal injury not observed in states of equivalent hypokalemia. *Kidney Int*. 2007;72(12):1483–1492.
71. Bakris GL, Weir MR. Angiotensin-converting enzyme inhibitor-associated elevations in serum creatinine: is this a cause for concern? *Arch Intern Med*. 2000;160(5):685–693.
72. Taler SJ, Agarwal R, Bakris GL, et al. KDOQI US commentary on the 2012 KDIGO clinical practice guideline for management of blood pressure in CKD. *Am J Kidney Dis*. 2013;62(2):201–213.
73. Qiao Y, Shin JI, Chen TK, et al. Association between renin-angiotensin system blockade discontinuation and all-cause mortality among persons with low estimated glomerular filtration rate. *JAMA Intern Med*. 2020;180(5):718–726.